

Diploma in Engineering and Technology • Semester 1

Applied Physics-I

Downloadable Notes

Student-friendly digital textbook + workbook + revision guide + answer guide

Subject Code: 1003

Credits: 3 • Periods/week: 3 • Periods/semester: 45

Includes bilingual explanations, formulas, SVG diagrams, solved numericals, practice questions, answer keys and long-answer hints.

How to use

Read one lesson, copy formulas, solve the worked examples, then test yourself using the answer key.

Notebook style

English + Malayalam

Exam practice

Printable PDF

Mobile friendly

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Course Overview

Course objectives

- Understand important physical principles used in engineering.
- Develop quantitative reasoning through numerical problems.
- Apply mechanics, energy, heat and fluid concepts to practical situations.

Course outcomes

- CO1: Apply laws of mechanics in rocket propulsion and recoil of gun.
- CO2: Apply concepts of circular and rotational motion.
- CO3: Solve problems involving work, energy, power, temperature and friction.
- CO4: Use fluid dynamics in atomiser and airfoil.

Course Roadmap



Module Map

Module 1: Measurements, Vectors and Laws of Motion

9 hours • CO1: Apply laws of mechanics in rocket propulsion and recoil of gun.

- Discuss physical quantities and units.
- Calculate errors in measurement.
- Differentiate scalar and vector quantities and add/resolve vectors.
- Apply Newton's laws and conservation of linear momentum.

Module 2: Circular Motion and Rotational Motion

9 hours • CO2: Apply concepts of circular motion and rotational motion.

- Explain angular displacement, angular velocity and angular acceleration.
- Apply circular motion to banking of roads and railway tracks.
- Understand moment of inertia and radius of gyration.
- Use torque and conservation of angular momentum.

Module 3: Work, Energy, Power, Friction and Heat

13 hours • CO3: Make use of energy, power, work, temperature and friction to solve problems.

- Explain work and friction.
- Use conservation of energy and transformations.
- Calculate energy and power in engineering situations.
- Understand heat, temperature scales, heat transfer and thermometry.

Module 4: Elasticity and Fluid Mechanics

12 hours • CO4: Use the theorems of fluid dynamics in atomiser and airfoil.

- Calculate stress, strain and elastic moduli.
- Explain pressure and surface tension.
- Understand viscosity, terminal velocity and Stoke's law.
- Apply continuity equation and Bernoulli theorem to atomiser and airfoil.

How to Study This PDF

Daily study method

1. Read the English explanation first.
2. Read the Malayalam explanation to confirm the idea.
3. Write all formulas with symbols and SI units.
4. Redraw the diagrams in your notebook.
5. Solve worked problems without seeing the answer.
6. Attempt practice problems and check the answer key.

Exam strategy

- For numerical problems: Given Data → Formula → Substitution → Answer with unit.
- For theory: Definition → Explanation → Diagram → Application → Example.
- For long answers: use headings and label diagrams.
- Revise formulas at least twice before the exam.

Note: Expected questions in this PDF are practice-focused and not a guarantee of actual examination questions.

Mini Formula Bank

$$v = s / t$$

Velocity

v: velocity, s: displacement, t: time

$$a = (v - u) / t$$

Acceleration

u: initial velocity, v: final velocity

$$F = ma$$

Force

m: mass, a: acceleration

$$p = mv$$

Momentum

SI unit: kg m s⁻¹

$$J = F t = \Delta p$$

Impulse

SI unit: N s

$$\omega = \theta / t$$

Angular velocity

SI unit: rad s⁻¹

$$v = r\omega$$

Linear-angular relation

r: radius

$$F = mv^2 / r$$

Centripetal force

Towards centre

$$W = F s \cos \theta$$

Work

SI unit: joule

$$KE = \frac{1}{2}mv^2$$

Kinetic energy

Energy due to motion

$$PE = mgh$$

Potential energy

Energy due to height

$$P = W / t$$

Power

SI unit: watt

$$\sigma = F / A$$

Stress

SI unit: Pa

$$\text{strain} = \Delta L / L$$

Strain

No unit

$$Y = \text{stress} / \text{strain}$$

Young modulus

SI unit: Pa

$$A_1 v_1 = A_2 v_2$$

Continuity

For incompressible fluid

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

Bernoulli

Ideal streamline flow

M1

Measurements, Vectors and Laws of Motion

CO1: Apply laws of mechanics in rocket propulsion and recoil of gun.

ഈ module-ൽ units, errors, vectors, Newton's laws, momentum, impulse, recoil of gun, rocket propulsion എന്നിവ exam point of view-ൽ പഠിക്കാം.

Learning outcomes:

- Discuss physical quantities and units.
- Calculate errors in measurement.
- Differentiate scalar and vector quantities and add/resolve vectors.
- Apply Newton's laws and conservation of linear momentum.

MODULE 1 • LESSON 1

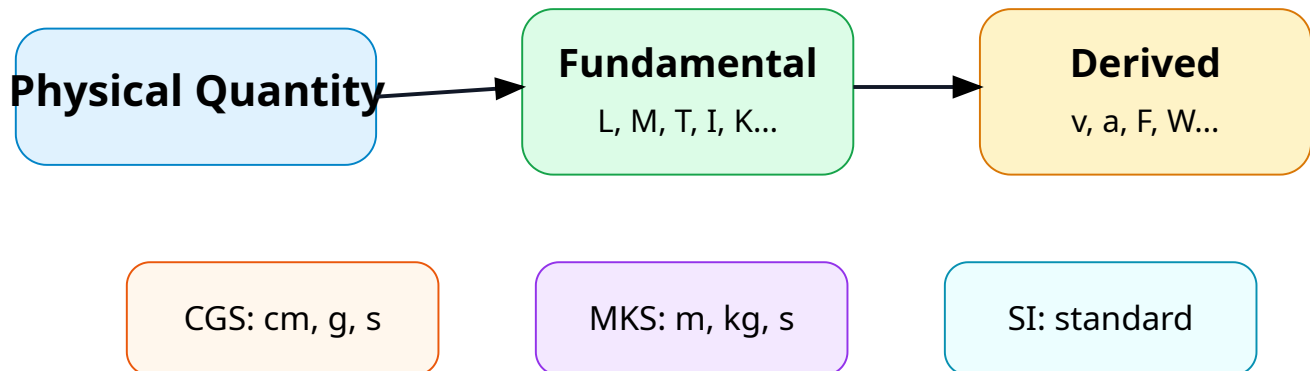
Physical quantities, units and systems

ENGLISH

A physical quantity is anything that can be measured and expressed with a number and a unit. Fundamental quantities are independent quantities such as length, mass and time. Derived quantities are formed by combining fundamental quantities, for example velocity, acceleration, force and work.

MALAYALAM

അളക്കാൻ കഴിയുന്ന ഒരു quantity ആണ് physical quantity. അതിന് ഒരു numerical value ഉം unit ഉം ഉണ്ടായിരിക്കും. Length, mass, time പോലുള്ളവ fundamental quantities ആണ്. Velocity, acceleration, force പോലുള്ളവ fundamental quantities ചേർത്ത് കിട്ടുന്ന derived quantities ആണ്.



Important formulas

Physical quantity
=
numerical value × unit

Dimension idea

Example: length
= 5 m

Symbols and SI units

Quantity	SI unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K

Real-life examples

- CGS uses centimetre, gram and second.
- MKS uses metre, kilogram and second.
- SI is the internationally accepted system for science and engineering.

Engineering applications

- Engineering drawings need accurate units.
- Machine design calculations use SI units to avoid conversion errors.

Study tip / memory trick: Number without unit is incomplete. Always write SI unit with the final answer.

Quick revision: Read the definition, formula, SI unit and one application. Most short-answer questions come from these four items.

MODULE 1 • LESSON 2

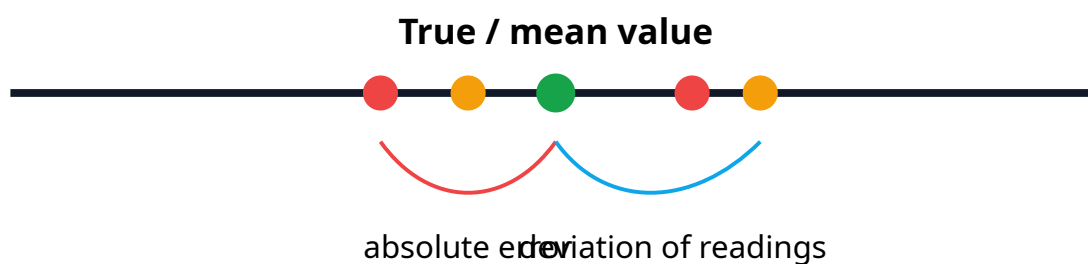
Errors in measurement

ENGLISH

Error is the difference between the measured value and the true or accepted value. Systematic error occurs due to faulty instruments or wrong method. Random error occurs due to unpredictable changes during repeated measurements.

MALAYALAM

Measured value-നും true/mean value-നും ഇടയിലുള്ള വ്യത്യാസമാണ് error. Instrument zero error അല്ലെങ്കിൽ wrong method കാരണം systematic error ഉണ്ടാകും. Repeated reading-കളിൽ ചെറിയ മാറ്റങ്ങൾ വന്നാൽ random error ഉണ്ടാകും.



Important formulas

absolute error =
|measured value - true value|

Absolute error
Magnitude only, no sign

relative error =
absolute error / true value

Relative error
No unit

% error =
relative error × 100

Percentage error
In percent

Symbols and SI units

Term	Meaning	Unit
Absolute error	Difference from true/mean value	same as measured quantity
Relative error	fractional error	no unit
Percentage error	relative error × 100	%

Real-life examples

- If a length is 2.48 cm instead of 2.50 cm, absolute error is 0.02 cm.
- Repeating an experiment and taking the mean reduces random error.

Engineering applications

- Laboratory experiments, quality inspection, machining tolerance and material testing.

Study tip / memory trick: Percentage error is useful because it compares accuracy of different measurements.

Quick revision: Read the definition, formula, SI unit and one application. Most short-answer questions come from these four items.

MODULE 1 • LESSON 3

Scalar and vector quantities

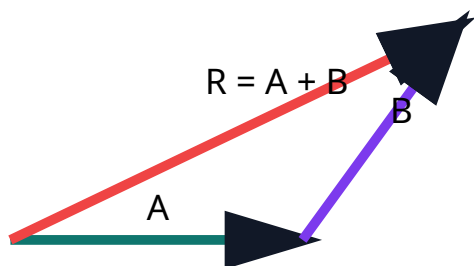
ENGLISH

A scalar has magnitude only; a vector has magnitude and direction. Temperature, mass and time are scalars. Displacement, velocity, acceleration, force and momentum are vectors. Vectors are represented by arrows where length shows magnitude and arrowhead shows direction.

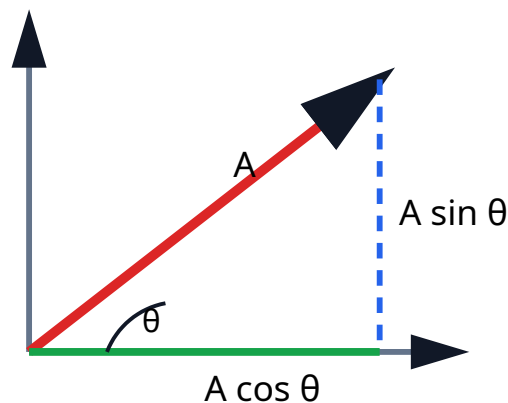
MALAYALAM

Scalar quantity-ക്ക് magnitude മാത്രം. Vector quantity-ക്ക് magnitude കൂടെ direction ഉം വേണം. Temperature, mass, time scalar ആണ്. Displacement, velocity, acceleration, force, momentum vector ആണ്. Arrow ഉപയോഗിച്ച് vector represent ചെയ്യുന്നു.

Triangle law



Resolution



Important formulas

$$R = \sqrt{(A^2 + B^2)}$$

Resultant of perpendicular vectors

When A and B are at 90°

$$A_x = A \cos \theta, A_y = A \sin \theta$$

Resolution

Horizontal and vertical components

Symbols and SI units

Symbol	Meaning	SI unit
R	resultant vector	depends on quantity
A_x	x-component	same as A
A_y	y-component	same as A
θ	angle with x-axis	degree or radian

Real-life examples

- Walking 3 m east and 4 m north gives displacement 5 m north-east.
- A force pulling a trolley at angle has horizontal and vertical components.

Engineering applications

- Bridge forces, crane lifting, slope motion, surveying, machine components.

Study tip / memory trick: For perpendicular vectors, think of a right triangle and use Pythagoras.

Quick revision: Read the definition, formula, SI unit and one application. Most short-answer questions come from these four items.

MODULE 1 • LESSON 4

Equations of motion and Newton's laws

ENGLISH

Equations of motion relate displacement, velocity, acceleration and time for uniformly accelerated motion. Newton's first law explains inertia, second law gives $F = ma$, and third law states action and reaction are equal and opposite.

MALAYALAM

Uniform acceleration ഉള്ള motion-ൽ displacement, velocity, acceleration, time തമ്മിലുള്ള ബന്ധം equations of motion പറയുന്നു. Newton's first law inertia explain ചെയ്യുന്നു. Second law $F = ma$. Third law: action-നും reaction-നും magnitude same, direction opposite.

Important formulas

$$v = u + at$$

First equation
final velocity

$$s = ut + \frac{1}{2}at^2$$

Second equation
displacement

$$v^2 = u^2 + 2as$$

Third equation
time not needed

$$F = ma$$

Newton's second law
Force causes acceleration

Symbols and SI units

Symbol	Meaning	SI unit
u	initial velocity	m s ⁻¹
v	final velocity	m s ⁻¹
a	acceleration	m s ⁻²
s	displacement	m
F	force	N

Real-life examples

- A bus starting from rest and gaining speed has acceleration.
- A passenger moves forward when a bus stops suddenly due to inertia.

Engineering applications

- Vehicle braking, elevator motion, conveyor design, machine dynamics.

Study tip / memory trick: Choose the equation that does not contain the unknown you do not need.

Quick revision: Read the definition, formula, SI unit and one application. Most short-answer questions come from these four items.

MODULE 1 • LESSON 5

Momentum, impulse, recoil and rocket propulsion

ENGLISH

Momentum is the quantity of motion of a body. Linear momentum is conserved when no external force acts on a system. Recoil of gun and rocket propulsion are important applications. Impulse is the product of force and short time interval and is equal to change in momentum.

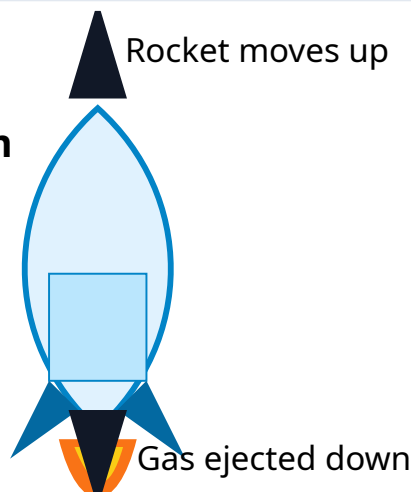
MALAYALAM

Momentum ഒരു body-യുടെ motion quantity ആണ്. External force ഇല്ലാത്ത system-ൽ total linear momentum constant ആയിരിക്കും. Gun recoil, rocket propulsion എന്നിവ conservation of momentum-ന്റെ applications ആണ്. Impulse = force × short time = change in momentum.

Conservation of momentum

Initial momentum = 0

Upward rocket momentum
= downward gas momentum



Important formulas

$$p = mv$$

Momentum
kg m s⁻¹

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

Conservation
closed system

$$J = Ft = \Delta p$$

Impulse
N s

Symbols and SI units

Symbol	Meaning	SI unit
p	momentum	kg m s ⁻¹
m	mass	kg
v	velocity	m s ⁻¹
J	impulse	N s
F	force	N
t	time	s

Real-life examples

- A cricket player lowers hands while catching to increase time and reduce force.
- A rocket moves forward because gases are expelled backward at high speed.

Engineering applications

- Rocket engines, jet propulsion, gun recoil control, impact safety design.

Study tip / memory trick: Impulse problems are usually easiest by using $J = \text{change in momentum}$.

Quick revision: Read the definition, formula, SI unit and one application. Most short-answer questions come from these four items.

Module 1 - Solved Numerical Problems

Every numerical follows the exam-friendly method: Given Data → Formula Used → Substitution → Calculation → Final Answer.

1. Error calculation from repeated readings

Given Data: Readings of length = 2.50 cm, 2.52 cm, 2.48 cm. Mean = $(2.50+2.52+2.48)/3 = 2.50$ cm.

Formula Used: absolute error = |reading - mean|, average absolute error = sum/errors ÷ n

Substitution: Errors = 0, 0.02, 0.02 cm

Calculation Steps: Average absolute error = $(0 + 0.02 + 0.02)/3 = 0.013$ cm. Relative error = $0.013/2.50 = 0.0052$.
Percentage error = 0.52%.

Final Answer: Average absolute error = 0.013 cm; percentage error = 0.52%

Shortcut Method: Take mean first; do not subtract from only one reading.

Common Mistakes: Ignoring units cm and writing negative error.

2. Resultant of two perpendicular vectors

Given Data: A = 3 N east, B = 4 N north.

Formula Used: $R = \sqrt{A^2+B^2}$, $\tan \theta = B/A$

Substitution: $R = \sqrt{3^2+4^2}$

Calculation Steps: $R = \sqrt{25} = 5$ N. $\tan \theta = 4/3$, $\theta = 53.1^\circ$ with east.

Final Answer: Resultant = 5 N at 53.1° north of east.

Shortcut Method: 3-4-5 triangle gives answer quickly.

Common Mistakes: Adding $3 + 4 = 7$ N without considering direction.

3. Force from Newton's second law

Given Data: Mass $m = 5$ kg, acceleration $a = 2 \text{ m s}^{-2}$.

Formula Used: $F = ma$

Substitution: $F = 5 \times 2$

Calculation Steps: $F = 10$ N.

Final Answer: Force = 10 N

Shortcut Method: When mass is in kg and acceleration in m/s^2 , answer is directly in newton.

Common Mistakes: Using gram instead of kg.

4. Recoil velocity of gun

Given Data: Bullet mass = 0.02 kg, bullet velocity = 500 m/s, gun mass = 5 kg. Initially system at rest.

Formula Used: $0 = m_b v_b + m_g v_g$

Substitution: $0 = 0.02 \times 500 + 5 v_g$

Calculation Steps: $5v_g = -10$, so $v_g = -2$ m/s. Negative sign means backward.

Final Answer: Gun recoils with speed 2 m/s backwards.

Shortcut Method: Recoil speed = bullet momentum ÷ gun mass.

Common Mistakes: Forgetting opposite direction sign.

5. Impulse due to a force

Given Data: Force $F = 50$ N acts for time $t = 0.2$ s.

Formula Used: $J = Ft$

Substitution: $J = 50 \times 0.2$

Calculation Steps: $J = 10$ N s. This is also the change in momentum.

Final Answer: Impulse = 10 N s

Shortcut Method: Multiply force by time directly.

Common Mistakes: Writing impulse unit as N only.

Module 1 - Practice Numericals with Answer Key

No.	Question	Answer / Hint
1	A body of mass 3 kg accelerates at 4 m/s^2 . Find force.	12 N
2	A 0.01 kg bullet moves at 600 m/s. Find momentum.	6 kg m/s
3	A force of 20 N acts for 0.5 s. Find impulse.	10 N s
4	Two perpendicular forces 6 N and 8 N act at a point. Find resultant.	10 N
5	Convert 50 cm to metre.	0.5 m
6	If true value is 100 g and measured value is 98 g, find percentage error.	2%
7	A 4 kg body changes velocity from 2 m/s to 8 m/s. Find change in momentum.	24 kg m/s
8	A car starts from rest and accelerates at 2 m/s^2 for 5 s. Find final velocity.	10 m/s

Numerical shortcut rule: Convert all quantities to SI units before substitution. Keep formula visible until the final step.

Module 1 - Exam-Oriented Questions and Answers

Short answer questions

Q1. What is a derived quantity?

Answer: A quantity obtained by combining fundamental quantities, for example velocity, acceleration or force.

Q2. Define absolute error.

Answer: The magnitude of the difference between measured value and true/mean value.

Q3. Give two examples of vector quantities.

Answer: Force and velocity.

Q4. State Newton's third law.

Answer: For every action there is an equal and opposite reaction.

Q5. What is impulse?

Answer: Impulse is the product of force and time and equals change in momentum.

Viva-style quick questions

Q1. Viva: What is a derived quantity?

Answer: A quantity obtained by combining fundamental quantities, for example velocity, acceleration or force.

Q2. Viva: Define absolute error.

Answer: The magnitude of the difference between measured value and true/mean value.

Q3. Viva: Give two examples of vector quantities.

Answer: Force and velocity.

Q4. Viva: State Newton's third law.

Answer: For every action there is an equal and opposite reaction.

Q5. Viva: What is impulse?

Answer: Impulse is the product of force and time and equals change in momentum.

Module 1 - Objective Practice with Answers

One-mark questions

Q1. SI unit of force is ____.

Answer: newton (N)

Q2. Momentum is a scalar/vector?

Answer: Vector

Q3. Percentage error = relative error $\times$ ____.

Answer: 100

Q4. Impulse equals change in ____.

Answer: momentum

Fill in the blanks

Q1. CGS unit of length is ____.

Answer: centimetre

Q2. A vector has magnitude and ____.

Answer: direction

Q3. The law used in rocket propulsion is conservation of ____.

Answer: linear momentum

Q4. $F = ma$ is Newton's ____ law.

Answer: second

True / False

Q1. Mass is a vector quantity.

Answer: False

Q2. Impulse has SI unit N s.

Answer: True

Q3. Random errors can be reduced by repeated readings.

Answer: True

Q4. Recoil velocity is in the same direction as bullet velocity.

Answer: False

Match the following

Item	Correct match
Scalar	Time
Vector	Force
Impulse	N s
Momentum	kg m/s

Assertion-Reason Questions

Q1. Assertion: A rocket can move in space. Reason: It uses conservation of momentum by ejecting gases backward.

Answer: Both assertion and reason are true, and reason correctly explains assertion.

Q2. Assertion: Work is done when a wall is pushed but it does not move. Reason: Work needs displacement.

Answer: Assertion is false; reason is true.

Module 1 - Long Answer Writing Practice

State and derive conservation of linear momentum. Explain recoil of gun as an application.

Key points to remember:

- Define momentum $p = mv$.
- Consider two bodies interacting without external force.
- Use Newton's third law: internal forces are equal and opposite.
- Show total momentum before collision = total momentum after collision.
- For gun and bullet initially at rest, total momentum is zero.
- Bullet gets forward momentum; gun gets equal backward momentum.

Important diagram to draw: Draw gun and bullet before and after firing with opposite arrows.

Mark-wise answer writing hint: For 10 marks: 2 marks statement, 4 marks derivation, 2 marks recoil explanation, 1 mark diagram, 1 mark units/sign convention.

Sample model answer: In an isolated system, when no external force acts, the total linear momentum remains constant. Let two bodies exert equal and opposite internal forces during interaction. Since force equals rate of change of momentum, the change in momentum of one body is equal and opposite to the change in momentum of the other body. Therefore the vector sum of momenta remains the same before and after interaction. In a gun-bullet system, initial momentum is zero. After firing, the bullet moves forward with momentum $m v$ and the gun moves backward with momentum $M V$. Hence $m v + M V = 0$, so the gun recoils in the backward direction.

1 mark

Write definition/formula with unit.

2 marks

Definition + one explanation or example.

5 marks

Statement + formula + diagram/application.

10 marks

Detailed explanation + derivation + diagram + application.

Module 1 - End-of-Module Revision Sheet

Quick formula cards

**Physical quantity =
numerical value × unit**

Dimension idea

Example: length = 5 m

**absolute error =
| measured value - true
value |**

Absolute error

Magnitude only, no sign

**relative error = absolute
error / true value**

Relative error

No unit

**% error = relative error ×
100**

Percentage error

In percent

$R = \sqrt{A^2 + B^2}$

Resultant of perpendicular vectors

When A and B are at 90°

$A_x = A \cos \theta$, $A_y = A \sin \theta$

Resolution

Horizontal and vertical components

$v = u + at$

First equation

final velocity

$s = ut + \frac{1}{2}at^2$

Second equation

displacement

$v^2 = u^2 + 2as$

Third equation

time not needed

$F = ma$

Newton's second law

Force causes acceleration

$p = mv$

Momentum

kg m s⁻¹

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

Conservation

closed system

Common mistakes to avoid

- Not writing SI units in final answer.
- Using formula without converting cm, mm or rpm to SI.
- Forgetting vector direction or sign.
- Skipping diagram in long answers.

Before exam checklist

- Can I define every key term?
- Can I write formulas from memory?
- Can I solve at least five numerical problems?
- Can I draw the important diagram neatly?

All exercise questions in this module include answers, answer keys, model answers or writing hints.

M2

Circular Motion and Rotational Motion

CO2: Apply concepts of circular motion and rotational motion.

ഈ module-ൽ circular motion, banking, moment of inertia, torque, angular momentum എന്നിവ പഠിക്കുന്നു.

Learning outcomes:

- Explain angular displacement, angular velocity and angular acceleration.
- Apply circular motion to banking of roads and railway tracks.
- Understand moment of inertia and radius of gyration.
- Use torque and conservation of angular momentum.

MODULE 2 • LESSON 1

Circular motion and angular quantities

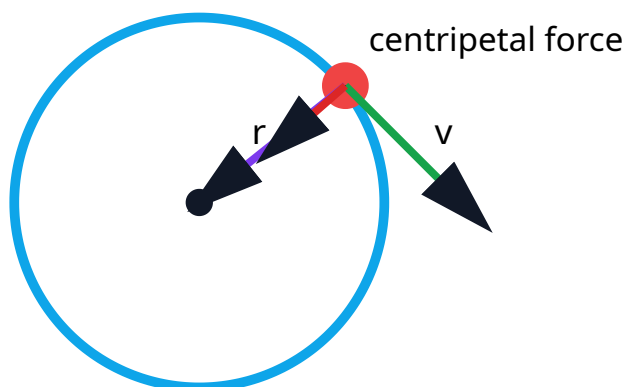
ENGLISH

Circular motion is motion along a circular path. Angular displacement is the angle swept by the radius vector. Angular velocity is the rate of angular displacement and angular acceleration is the rate of change of angular velocity.

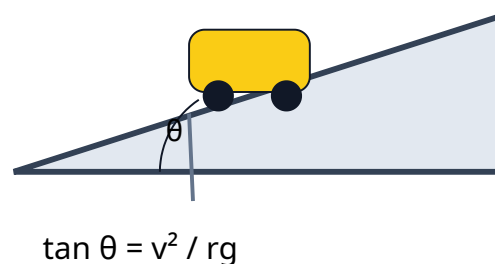
MALAYALAM

Circular path-ൽ നടക്കുന്ന motion ആണ് circular motion. Radius vector sweep ചെയ്യുന്ന angle ആണ് angular displacement. Angular displacement per second ആണ് angular velocity. Angular velocity മാറുന്ന rate ആണ് angular acceleration.

Circular motion



Banking of road



Important formulas

$$\omega = \theta/t$$

Angular velocity
rad s⁻¹

$$\alpha = (\omega - \omega_0)/t$$

Angular acceleration
rad s⁻²

$$v = r\omega$$

Linear velocity
m s⁻¹

$$a = r\alpha$$

Linear acceleration
m s⁻²

Symbols and SI units

Symbol	Meaning	SI unit
θ	angular displacement	radian
ω	angular velocity	rad s ⁻¹
α	angular acceleration	rad s ⁻²
r	radius	m
v	linear velocity	m s ⁻¹

Real-life examples

- A wheel rotating about its axle.
- A fan blade tip has circular motion.

Engineering applications

- Machine shafts, pulleys, wheels, gears, fans and turbines.

Study tip / memory trick: Linear quantity = radius × angular quantity.

Quick revision: Read the definition, formula, SI unit and one application. Most short-answer questions come from these four items.

MODULE 2 • LESSON 2

Centripetal acceleration, force and banking

ENGLISH

In circular motion, velocity direction continuously changes. Therefore an acceleration acts towards the centre called centripetal acceleration. The force producing it is centripetal force. Banking of roads provides a horizontal component of normal reaction to help vehicles turn safely.

MALAYALAM

Circular motion-ൽ speed constant ആയാലും velocity direction മാറുന്നു. അതിനാൽ centre-ലേക്ക് acceleration ഉണ്ടാകും; അതാണ് centripetal acceleration. അതിന് ആവശ്യമായ force centripetal force ആണ്. Banking road-ൽ normal reaction-ന്റെ horizontal component turning-ന് സഹായിക്കുന്നു.

Important formulas

$$a_c = \frac{v^2}{r} = r\omega^2$$

Centripetal acceleration
towards centre

$$F_c = \frac{mv^2}{r}$$

Centripetal force
newton

$$\tan \theta = \frac{v^2}{rg}$$

Banking angle
friction neglected

Symbols and SI units

Symbol	Meaning	SI unit
a_c	centripetal acceleration	m s^{-2}
F_c	centripetal force	N
θ	angle of banking	degree
g	acceleration due to gravity	m s^{-2}

Real-life examples

- A stone tied to a string and whirled in a circle.
- Vehicles slow down on sharp curves to reduce required centripetal force.

Engineering applications

- Highway curves, railway tracks, roller coasters and centrifuges.

Study tip / memory trick: Sharp curve means smaller r ; smaller r needs larger centripetal force.

Quick revision: Read the definition, formula, SI unit and one application. Most short-answer questions come from these four items.

MODULE 2 • LESSON 3

Moment of inertia and radius of gyration

ENGLISH

Moment of inertia is the rotational analogue of mass. It measures how difficult it is to change rotational motion. It

MALAYALAM

Moment of inertia rotational motion-ൽ mass പോലെയാണ്. Rotation മാറ്റാൻ എത്ര difficulty ഉണ്ടെന്ന്

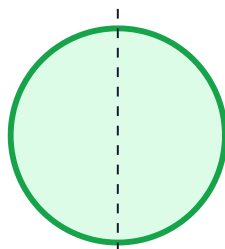
depends on mass and distribution of mass from the axis of rotation. Radius of gyration is the distance at which the whole mass may be assumed concentrated to give the same moment of inertia.

ഇത് കാണിക്കുന്നു. Mass എങ്ങനെ axis-ൽ നിന്ന് distribute ചെയ്തിരിക്കുന്നു എന്നതിനെ ആശ്രയിക്കും. Radius of gyration എന്നത് same moment of inertia ലഭിക്കാൻ മുഴുവൻ mass concentrated ആയി assume ചെയ്യാവുന്ന distance ആണ്.

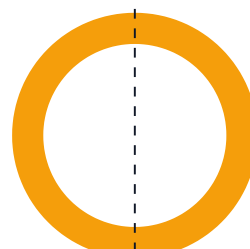
Moment of inertia depends on mass distribution



Rod: $I = ML^2/12$



Disc: $I = MR^2/2$



Ring: $I = MR^2$

Important formulas

$$I = \Sigma mr^2$$

Definition
kg m²

$$I = Mk^2$$

Radius of gyration
k in metre

$$I = I_G + Md^2$$

Parallel axis theorem
axes parallel

$$I_z = I_x + I_y$$

Perpendicular axis theorem
plane lamina

Symbols and SI units

Symbol	Meaning	SI unit
I	moment of inertia	kg m ²
M	total mass	kg
k	radius of gyration	m
d	distance between axes	m

Real-life examples

- A flywheel stores rotational energy because of large moment of inertia.
- A ring has more moment of inertia than a disc of same mass and radius.

Engineering applications

- Flywheels, gears, turbine rotors, shafts and rotating machine design.

Study tip / memory trick: Mass far from axis gives larger moment of inertia.

Quick revision: Read the definition, formula, SI unit and one application. Most short-answer questions come from these four items.

MODULE 2 • LESSON 4

Torque and angular momentum

ENGLISH

Torque is the turning effect of a force. Angular momentum is the rotational analogue of linear momentum. If no external torque acts on a system, angular momentum remains conserved.

MALAYALAM

Force-ന്റെ turning effect ആണ് torque. Angular momentum rotational motion-ലെ momentum പോലെ ആണ്. External torque ഇല്ലെങ്കിൽ angular momentum constant ആയിരിക്കും.

Important formulas

$$\tau = F \times r = Fr \sin \theta$$

Torque
N m

$$L = I\omega$$

Angular momentum
kg m² s⁻¹

$$I_1\omega_1 = I_2\omega_2$$

Conservation
no external torque

Symbols and SI units

Symbol	Meaning	SI unit
τ	torque	N m
L	angular momentum	kg m ² s ⁻¹
I	moment of inertia	kg m ²
ω	angular velocity	rad s ⁻¹

Real-life examples

- A longer spanner gives more torque.
- A skater spins faster when arms are pulled inward.

Engineering applications

- Spanners, steering wheels, engines, turbines, flywheels and gyroscopes.

Study tip / memory trick: Torque needs force and perpendicular distance from axis.

Quick revision: Read the definition, formula, SI unit and one application. Most short-answer questions come from these four items.

Module 2 - Solved Numerical Problems

Every numerical follows the exam-friendly method: Given Data → Formula Used → Substitution → Calculation → Final Answer.

1. Convert rpm to angular velocity and find linear velocity

Given Data: Wheel rotates at 120 rpm. Radius $r = 0.25$ m.

Formula Used: $\omega = 2\pi N/60$, $v = r\omega$

Substitution: $\omega = 2\pi \times 120/60 = 4\pi$ rad/s

Calculation Steps: $v = 0.25 \times 4\pi = \pi = 3.14$ m/s.

Final Answer: Angular velocity = 12.57 rad/s; linear velocity = 3.14 m/s

Shortcut Method: N rpm $\rightarrow \omega = \pi N/30$.

Common Mistakes: Using rpm directly without converting to per second.

2. Centripetal force on a car

Given Data: Mass $m = 1000$ kg, speed $v = 20$ m/s, radius $r = 80$ m.

Formula Used: $F = mv^2/r$

Substitution: $F = 1000 \times 20^2/80$

Calculation Steps: $F = 1000 \times 400/80 = 5000$ N.

Final Answer: Centripetal force = 5000 N

Shortcut Method: Square speed first.

Common Mistakes: Forgetting that v^2 means 20×20 .

3. Banking angle

Given Data: Speed $v = 20$ m/s, radius $r = 50$ m, $g = 9.8$ m/s².

Formula Used: $\tan \theta = v^2/(rg)$

Substitution: $\tan \theta = 400/(50 \times 9.8)$

Calculation Steps: $\tan \theta = 400/490 = 0.816$. $\theta = \tan^{-1}(0.816) = 39.2^\circ$.

Final Answer: Banking angle $\approx 39.2^\circ$

Shortcut Method: Use calculator tan inverse.

Common Mistakes: Keeping g as 10 may give approximate answer; state if approximation used.

4. Moment of inertia of ring

Given Data: Mass $M = 2$ kg, radius $R = 0.5$ m.

Formula Used: $I = MR^2$

Substitution: $I = 2 \times (0.5)^2$

Calculation Steps: $I = 2 \times 0.25 = 0.5$ kg m².

Final Answer: Moment of inertia = 0.5 kg m²

Shortcut Method: For ring about central axis use MR^2 directly.

Common Mistakes: Using disc formula by mistake.

5. Torque by a force

Given Data: Force $F = 20$ N, perpendicular distance $r = 0.3$ m.

Formula Used: $\tau = Fr$

Substitution: $\tau = 20 \times 0.3$

Calculation Steps: $\tau = 6$ N m.

Final Answer: Torque = 6 N m

Shortcut Method: If force is perpendicular, $\sin \theta = 1$.

Common Mistakes: Writing torque unit as N/m.

Module 2 - Practice Numericals with Answer Key

No.	Question	Answer / Hint
1	A wheel has angular velocity 10 rad/s and radius 0.2 m. Find linear velocity.	2 m/s
2	Find centripetal acceleration for $v = 15$ m/s and $r = 45$ m.	5 m/s ²
3	Mass 2 kg moves in circle of radius 1 m with speed 4 m/s. Find centripetal force.	32 N
4	Find angular momentum if $I = 0.4$ kg m ² and $\omega = 5$ rad/s.	2 kg m ² /s
5	A torque of 8 N m acts on a body. If force is 20 N, find arm length.	0.4 m
6	For a body $I = 6$ kg m ² and $M = 24$ kg. Find radius of gyration.	0.5 m
7	A skater changes I from 4 to 2 kg m ² . If initial $\omega = 3$ rad/s, find final ω .	6 rad/s
8	A road curve has $r = 100$ m and $v = 10$ m/s. Find $\tan \theta$. Use $g = 10$.	0.1

Numerical shortcut rule: Convert all quantities to SI units before substitution. Keep formula visible until the final step.

Module 2 - Exam-Oriented Questions and Answers

Short answer questions

Q1. Define angular velocity.

Answer: Angular velocity is the rate of change of angular displacement.

Q2. What is centripetal force?

Answer: The force directed towards the centre that keeps a body in circular motion.

Q3. Why are roads banked?

Answer: To provide required centripetal force and reduce skidding at curves.

Q4. What is moment of inertia?

Answer: It is the measure of resistance offered by a body to change in rotational motion.

Q5. State conservation of angular momentum.

Answer: If no external torque acts, total angular momentum remains constant.

Viva-style quick questions

Q1. Viva: Define angular velocity.

Answer: Angular velocity is the rate of change of angular displacement.

Q2. Viva: What is centripetal force?

Answer: The force directed towards the centre that keeps a body in circular motion.

Q3. Viva: Why are roads banked?

Answer: To provide required centripetal force and reduce skidding at curves.

Q4. Viva: What is moment of inertia?

Answer: It is the measure of resistance offered by a body to change in rotational motion.

Q5. Viva: State conservation of angular momentum.

Answer: If no external torque acts, total angular momentum remains constant.

Module 2 - Objective Practice with Answers

One-mark questions

Q1. Unit of angular velocity is ____.

Answer: rad/s

Q2. Centripetal force acts towards the ____.

Answer: centre

Q3. Moment of inertia SI unit is ____.

Answer: kg m²

Q4. Torque = force × ____.

Answer: perpendicular distance

Fill in the blanks

Q1. $v = r$ ____.

Answer: ω

Q2. For banking, $\tan \theta =$ ____.

Answer: v^2/rg

Q3. Angular momentum $L =$ ____.

Answer: $I\omega$

Q4. Radius of gyration symbol is ____.

Answer: k

True / False

Q1. Centripetal acceleration is directed away from centre.

Answer: False

Q2. A ring has $I = MR^2$ about central axis.

Answer: True

Q3. Angular momentum is conserved if external torque is zero.

Answer: True

Q4. Torque has unit joule.

Answer: False; torque unit is N m

Match the following

Item	Correct match
Torque	N m
Angular momentum	kg m ² /s
Centripetal force	mv^2/r
Banking	v^2/rg

Assertion-Reason Questions

Q1. Assertion: Roads are banked at curves. Reason: Banking helps to provide centripetal force.

Answer: Both true and reason explains assertion.

Q2. Assertion: Moment of inertia depends only on mass. Reason: Distribution of mass from axis is important.

Answer: Assertion false; reason true.

Module 2 - Long Answer Writing Practice

Explain banking of roads and derive the formula $\tan \theta = v^2/rg$ for a frictionless banked road.

Key points to remember:

- Draw a vehicle on an inclined road surface.
- Show weight mg vertically downward and normal reaction R perpendicular to road.
- Resolve R into vertical and horizontal components.
- Use $R \cos \theta = mg$ for vertical balance.
- Use $R \sin \theta = mv^2/r$ for centripetal force.
- Divide the two equations to get $\tan \theta = v^2/rg$.

Important diagram to draw: Draw a car on an inclined road with R , mg , θ and centripetal direction towards centre.

Mark-wise answer writing hint: For 10 marks, diagram is essential. Label forces clearly and mention friction neglected.

Sample model answer: On a banked road, the road surface is inclined at an angle θ to the horizontal. The normal reaction R acts perpendicular to the surface and weight mg acts vertically downward. The vertical component $R \cos \theta$ balances weight, so $R \cos \theta = mg$. The horizontal component $R \sin \theta$ provides centripetal force, so $R \sin \theta = mv^2/r$. Dividing the second equation by the first gives $\tan \theta = v^2/rg$. Thus banking helps vehicles take turns safely.

1 mark

Write definition/formula with unit.

2 marks

Definition + one explanation or example.

5 marks

Statement + formula + diagram/application.

10 marks

Detailed explanation + derivation + diagram + application.

Module 2 - End-of-Module Revision Sheet

Quick formula cards

$$\omega = \theta/t$$

Angular velocity

rad s⁻¹

$$\alpha = (\omega - \omega_0)/t$$

Angular acceleration

rad s⁻²

$$v = r\omega$$

Linear velocity

m s⁻¹

$$a = r\alpha$$

Linear acceleration

m s⁻²

$$a_c = v^2/r = r\omega^2$$

Centripetal acceleration

towards centre

$$F_c = mv^2/r$$

Centripetal force

newton

$$\tan \theta = v^2/(rg)$$

Banking angle

friction neglected

$$I = \Sigma mr^2$$

Definition

kg m²

$$I = Mk^2$$

Radius of gyration

k in metre

$$I = I_G + Md^2$$

Parallel axis theorem

axes parallel

$$I_z = I_x + I_y$$

Perpendicular axis theorem

plane lamina

$$\tau = F \times r = Fr \sin \theta$$

Torque

N m

Common mistakes to avoid

- Not writing SI units in final answer.
- Using formula without converting cm, mm or rpm to SI.
- Forgetting vector direction or sign.
- Skipping diagram in long answers.

Before exam checklist

- Can I define every key term?
- Can I write formulas from memory?
- Can I solve at least five numerical problems?
- Can I draw the important diagram neatly?

All exercise questions in this module include answers, answer keys, model answers or writing hints.

M3

Work, Energy, Power, Friction and Heat

CO3: Make use of energy, power, work, temperature and friction to solve problems.

ഈ module-ൽ work, friction, energy, power, heat, temperature scales, heat transfer, thermometer, pyrometer എന്നിവ പഠിക്കുന്നു.

Learning outcomes:

- Explain work and friction.
- Use conservation of energy and transformations.
- Calculate energy and power in engineering situations.
- Understand heat, temperature scales, heat transfer and thermometry.

MODULE 3 • LESSON 1

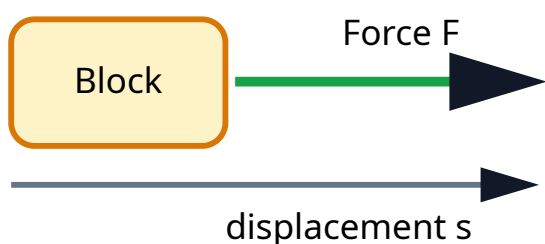
Work and friction

ENGLISH

Work is done when a force produces displacement. Work depends on force, displacement and angle between them. Friction is a resistive force that opposes relative motion. Static friction acts before motion starts and kinetic friction acts during motion.

MALAYALAM

Force കാരണം displacement ഉണ്ടാകുമ്പോൾ work done എന്ന് പറയുന്നു. Work force, displacement, angle എന്നിവയെ ആശ്രയിക്കും. Relative motion-നെ oppose ചെയ്യുന്ന resistive force ആണ് friction. Motion start ചെയ്യുന്നതിന് മുൻപ് static friction, motion നടക്കുമ്പോൾ kinetic friction.



Energy transformation



Falling body: PE decreases, KE inc

Important formulas

$$W = Fs \cos \theta$$

Work
joule

$$F = \mu N$$

Limiting friction
 μ : coefficient of friction

$$N = mg$$

On horizontal plane
normal reaction

Symbols and SI units

Symbol	Meaning	SI unit
W	work	J
F	force	N
s	displacement	m
μ	coefficient of friction	no unit
N	normal reaction	N

Real-life examples

- Pushing a trolley through a distance.
- Friction helps walking but wastes energy in machines.

Engineering applications

- Brake systems, belt drives, clutches, bearings, machine lubrication.

Study tip / memory trick: No displacement means no mechanical work even if force is applied.

Quick revision: Read the definition, formula, SI unit and one application. Most short-answer questions come from these four items.

MODULE 3 • LESSON 2

Energy, power and conservation of energy

ENGLISH

Energy is the capacity to do work. Kinetic energy is due to motion and potential energy is due to position or height. According to conservation of energy, energy cannot be created or destroyed; it only changes from one form to another.

MALAYALAM

Work ചെയ്യാനുള്ള capacity ആണ് energy. Motion കാരണം ഉള്ള energy kinetic energy. Height/position കാരണം ഉള്ള energy potential energy. Conservation of energy പ്രകാരം energy create ചെയ്യാനും destroy ചെയ്യാനും കഴിയില്ല; അത് form മാത്രം മാറും.

Important formulas

$$KE = \frac{1}{2}mv^2$$

Kinetic energy
joule

$$PE = mgh$$

Potential energy
joule

$$P = W/t = E/t$$

Power
watt

$$1 \text{ hp} = 746 \text{ W}$$

Horsepower
engineering use

Symbols and SI units

Symbol	Meaning	SI unit
KE	kinetic energy	J
PE	potential energy	J
P	power	W
g	acceleration due to gravity	m s^{-2}
h	height	m

Real-life examples

- A falling stone converts PE into KE.
- An electric motor converts electrical energy to mechanical energy.

Engineering applications

- Lifts, motors, cranes, hydroelectric power plants and vehicle engines.

Study tip / memory trick: Power tells how fast work is done.

Quick revision: Read the definition, formula, SI unit and one application. Most short-answer questions come from these four items.

MODULE 3 • LESSON 3

Heat, temperature and temperature scales

ENGLISH

MALAYALAM

Heat is energy transferred due to temperature difference. Temperature measures hotness or coldness of a body. Common scales are Celsius, Fahrenheit and Kelvin. Kelvin is the SI unit of temperature.

Temperature difference കാരണം transfer ചെയ്യുന്ന energy ആണ് heat. ഒരു body എത്ര hot/cold ആണെന്ന് കാണിക്കുന്നത് temperature ആണ്. Celsius, Fahrenheit, Kelvin scales ഉപയോഗിക്കുന്നു. Temperature-ന്റെ SI unit Kelvin ആണ്.

Important formulas

$$K = ^\circ C + 273$$

Celsius to Kelvin
approximate for diploma problems

$$F = (9/5)C + 32$$

Celsius to Fahrenheit
temperature conversion

$$C/5 = (F - 32)/9 = (K - 273)/5$$

Scale relation
common relation

$$Q = mc\Delta T$$

Heat gained
specific heat problems

Symbols and SI units

Symbol	Meaning	SI unit
Q	heat energy	J
m	mass	kg
c	specific heat	J kg ⁻¹ K ⁻¹
ΔT	change in temperature	K or $^\circ C$

Real-life examples

- A hot tea cup cools because heat flows to surroundings.
- A pyrometer measures high temperature without touching the object.

Engineering applications

- Furnaces, boilers, heat treatment, thermal measurement and HVAC.

Study tip / memory trick: Temperature difference in $^\circ C$ and K has the same numerical value.

Quick revision: Read the definition, formula, SI unit and one application. Most short-answer questions come from these four items.

MODULE 3 • LESSON 4

Heat transfer, specific heat, thermometer and pyrometer

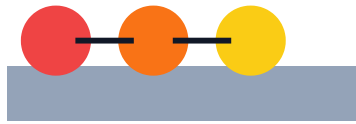
ENGLISH

Heat is transferred by conduction, convection and radiation. Conduction occurs through solids mainly, convection occurs through fluid motion and radiation needs no material medium. Specific heat is the heat required to raise the temperature of unit mass by one kelvin.

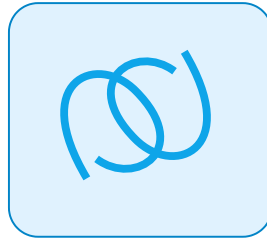
MALAYALAM

Heat transfer conduction, convection, radiation വഴിയാണ്. Solids-ൽ conduction പ്രധാനമാണ്. Fluids-ൽ convection. Radiation-ന് medium ആവശ്യമില്ല. Unit mass-ന്റെ temperature 1 K ഉയർത്താൻ വേണ്ട heat ആണ് specific heat.

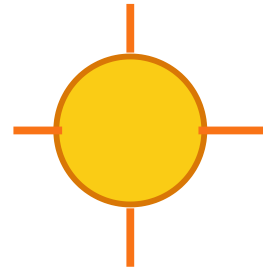
Modes of heat transfer



Conduction



Convection



Radiation

Important formulas

$$Q = mc\Delta T$$

Specific heat
relation
J

$$F = 1.8C + 32$$

Temperature
conversion
°F

Symbols and SI units

Instrument	Use	Important point
Mercury thermometer	ordinary temperature measurement	mercury expands uniformly
Pyrometer	high temperature measurement	non-contact method

Real-life examples

- Metal spoon gets hot by conduction.
- Sea breeze is due to convection.
- Sun heat reaches Earth by radiation.

Engineering applications

- Insulation, heat exchangers, furnaces, boilers, engine cooling systems.

Study tip / memory trick: Conduction needs contact; radiation can travel through vacuum.

Quick revision: Read the definition, formula, SI unit and one application. Most short-answer questions come from these four items.

Module 3 - Solved Numerical Problems

Every numerical follows the exam-friendly method: Given Data → Formula Used → Substitution → Calculation → Final Answer.

1. Work done on horizontal plane

Given Data: Force $F = 100 \text{ N}$, displacement $s = 5 \text{ m}$, force along displacement.

Formula Used: $W = Fs$

Substitution: $W = 100 \times 5$

Calculation Steps: $W = 500 \text{ J}$.

Final Answer: Work done = 500 J

Shortcut Method: If $\theta = 0^\circ$, $\cos \theta = 1$.

Common Mistakes: Using kg instead of newton for force.

2. Frictional force

Given Data: Mass $m = 50 \text{ kg}$, coefficient $\mu = 0.4$, $g = 9.8 \text{ m/s}^2$.

Formula Used: $F = \mu N$, $N = mg$

Substitution: $F = 0.4 \times 50 \times 9.8$

Calculation Steps: $F = 196 \text{ N}$.

Final Answer: Limiting friction = 196 N

Shortcut Method: On horizontal plane, use $N = mg$.

Common Mistakes: Forgetting normal reaction.

3. Kinetic energy

Given Data: Mass $m = 10 \text{ kg}$, velocity $v = 4 \text{ m/s}$.

Formula Used: $KE = \frac{1}{2}mv^2$

Substitution: $KE = \frac{1}{2} \times 10 \times 4^2$

Calculation Steps: $KE = 5 \times 16 = 80 \text{ J}$.

Final Answer: Kinetic energy = 80 J

Shortcut Method: Square velocity first, then multiply by half mass.

Common Mistakes: Not squaring velocity.

4. Potential energy

Given Data: Mass $m = 5$ kg, height $h = 2$ m, $g = 9.8$ m/s².

Formula Used: $PE = mgh$

Substitution: $PE = 5 \times 9.8 \times 2$

Calculation Steps: $PE = 98$ J.

Final Answer: Potential energy = 98 J

Shortcut Method: Use $g = 9.8$ unless problem says 10.

Common Mistakes: Missing height unit metre.

5. Temperature conversion

Given Data: Celsius temperature $C = 25^\circ\text{C}$.

Formula Used: $F = (9/5)C + 32$, $K = C + 273$

Substitution: $F = 1.8 \times 25 + 32$, $K = 25 + 273$

Calculation Steps: $F = 45 + 32 = 77^\circ\text{F}$. $K = 298$ K.

Final Answer: $25^\circ\text{C} = 77^\circ\text{F} = 298$ K

Shortcut Method: Remember room temperature $25^\circ\text{C} \approx 298$ K.

Common Mistakes: Writing $^\circ\text{K}$; correct unit is K.

6. Heat required

Given Data: Mass $m = 2$ kg, $c = 500$ J kg⁻¹ K⁻¹, $\Delta T = 10$ K.

Formula Used: $Q = mc\Delta T$

Substitution: $Q = 2 \times 500 \times 10$

Calculation Steps: $Q = 10000$ J.

Final Answer: Heat required = 10000 J

Shortcut Method: Temperature rise of 10°C is same as 10 K.

Common Mistakes: Using final temperature instead of temperature change.

Module 3 - Practice Numericals with Answer Key

No.	Question	Answer / Hint
1	A force of 40 N moves a body through 3 m. Find work.	120 J
2	Find KE of a 2 kg body moving at 5 m/s.	25 J
3	Find PE of 10 kg at 4 m height. Use $g = 10$.	400 J
4	A motor does 2000 J work in 20 s. Find power.	100 W
5	Convert 100°C to Fahrenheit.	212°F
6	Convert 300 K to Celsius approximately.	27°C
7	Find heat needed for $m = 1$ kg, $c = 4200$ J/kgK, $\Delta T = 5$ K.	21000 J
8	If $\mu = 0.2$ and normal reaction = 100 N, find friction.	20 N

Numerical shortcut rule: Convert all quantities to SI units before substitution. Keep formula visible until the final step.

Module 3 - Exam-Oriented Questions and Answers

Short answer questions

Q1. Define work.

Answer: Work is done when a force produces displacement; $W = Fs \cos \theta$.

Q2. What is kinetic friction?

Answer: Friction acting when two surfaces are in relative motion.

Q3. State conservation of energy.

Answer: Energy cannot be created or destroyed; it transforms from one form to another.

Q4. What is power?

Answer: Power is the rate of doing work.

Q5. Name three modes of heat transfer.

Answer: Conduction, convection and radiation.

Viva-style quick questions

Q1. Viva: Define work.

Answer: Work is done when a force produces displacement; $W = Fs \cos \theta$.

Q2. Viva: What is kinetic friction?

Answer: Friction acting when two surfaces are in relative motion.

Q3. Viva: State conservation of energy.

Answer: Energy cannot be created or destroyed; it transforms from one form to another.

Q4. Viva: What is power?

Answer: Power is the rate of doing work.

Q5. Viva: Name three modes of heat transfer.

Answer: Conduction, convection and radiation.

Module 3 - Objective Practice with Answers

One-mark questions

Q1. SI unit of work is ____.

Answer: joule

Q2. Energy due to motion is ____.

Answer: kinetic energy

Q3. Power is rate of doing ____.

Answer: work

Q4. SI unit of temperature is ____.

Answer: kelvin

Fill in the blanks

Q1. $KE =$ ____.

Answer: $\frac{1}{2}mv^2$

Q2. $PE =$ ____.

Answer: mgh

Q3. 1 hp = ____ W.

Answer: 746

Q4. Heat transfer through vacuum is ____.

Answer: radiation

True / False

Q1. Static friction acts before motion starts.

Answer: True

Q2. Power has unit joule.

Answer: False; unit is watt

Q3. $Kelvin = Celsius + 273$ approximately.

Answer: True

Q4. Convection occurs mainly in solids.

Answer: False

Match the following

Item	Correct match
Work	J
Power	W
Specific heat	J/kgK
Conduction	solids

Assertion-Reason Questions

Q1. Assertion: A metal spoon in hot tea becomes hot. Reason: Heat transfers through conduction.

Answer: Both true and reason explains assertion.

Q2. Assertion: Radiation needs a material medium. Reason: Sunlight reaches Earth through vacuum.

Answer: Assertion false; reason true.

Module 3 - Long Answer Writing Practice

Explain conservation of energy for a freely falling body.

Key points to remember:

- Consider a body of mass m falling from height h .
- At top: $PE = mgh$, $KE = 0$.
- During fall: height decreases, speed increases.
- PE decreases and KE increases.
- At ground: $PE = 0$, $KE = mgh$ if air resistance is neglected.
- Total mechanical energy remains constant.

Important diagram to draw: Draw three positions: top, middle and bottom, with PE and KE labels.

Mark-wise answer writing hint: For full marks, clearly state that air resistance is neglected and total $PE + KE$ is constant.

Sample model answer: When a body falls freely from a height h , only gravity is considered and air resistance is neglected. At the highest point, velocity is zero, so kinetic energy is zero and potential energy is mgh . As the body falls, its height decreases, so potential energy decreases. At the same time velocity increases, so kinetic energy increases. At any intermediate point the sum of kinetic energy and potential energy remains equal to the initial value mgh . Just before reaching the ground, potential energy becomes zero and kinetic energy becomes maximum. Thus energy is transformed from potential to kinetic form without loss.

1 mark

Write definition/formula with unit.

2 marks

Definition + one explanation or example.

5 marks

Statement + formula + diagram/application.

10 marks

Detailed explanation + derivation + diagram + application.

Module 3 - End-of-Module Revision Sheet

Quick formula cards

$$W = Fs \cos \theta$$

Work

joule

$$F = \mu N$$

Limiting friction

μ : coefficient of friction

$$N = mg$$

On horizontal plane

normal reaction

$$KE = \frac{1}{2}mv^2$$

Kinetic energy

joule

$$PE = mgh$$

Potential energy

joule

$$P = W/t = E/t$$

Power

watt

$$1 \text{ hp} = 746 \text{ W}$$

Horsepower

engineering use

$$K = ^\circ\text{C} + 273$$

Celsius to Kelvin

approximate for diploma problems

$$F = (9/5)C + 32$$

Celsius to Fahrenheit

temperature conversion

$$C/5 = (F - 32)/9 = (K - 273)/5$$

Scale relation

common relation

$$Q = mc\Delta T$$

Heat gained

specific heat problems

$$Q = mc\Delta T$$

Specific heat relation

J

Common mistakes to avoid

- Not writing SI units in final answer.
- Using formula without converting cm, mm or rpm to SI.
- Forgetting vector direction or sign.
- Skipping diagram in long answers.

Before exam checklist

- Can I define every key term?
- Can I write formulas from memory?
- Can I solve at least five numerical problems?
- Can I draw the important diagram neatly?

All exercise questions in this module include answers, answer keys, model answers or writing hints.

M4

Elasticity and Fluid Mechanics

CO4: Use the theorems of fluid dynamics in atomiser and airfoil.

ഈ module-ൽ elasticity, pressure, surface tension, viscosity, continuity equation, Bernoulli theorem, airfoil, atomiser എന്നിവ പഠിക്കുന്നു.

Learning outcomes:

- Calculate stress, strain and elastic moduli.
- Explain pressure and surface tension.
- Understand viscosity, terminal velocity and Stoke's law.
- Apply continuity equation and Bernoulli theorem to atomiser and airfoil.

MODULE 4 • LESSON 1

Elasticity, stress, strain and Hooke's law

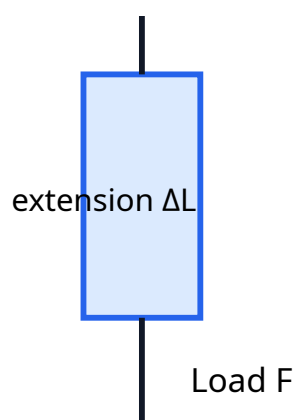
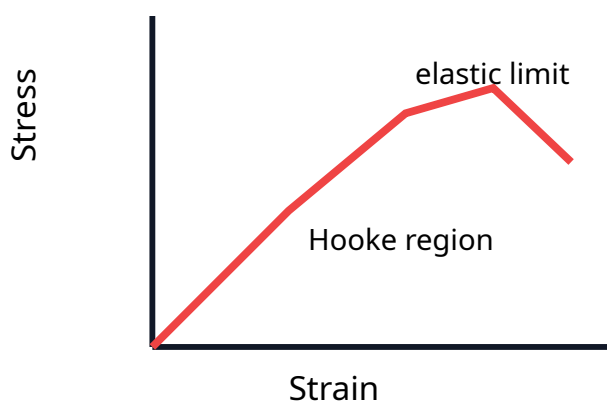
ENGLISH

Elasticity is the property by which a body regains its original shape and size after removal of deforming force. Stress is internal restoring force per unit area. Strain is the ratio of change in dimension to original dimension. Hooke's law states that, within elastic limit, stress is directly proportional to strain.

MALAYALAM

Deforming force remove ചെയ്താൽ original shape/size തിരിച്ചുപിടിക്കുന്ന property ആണ് elasticity. Unit area-യിലെ internal restoring force ആണ് stress. Original dimension-നോട് change in dimension-ന്റെ ratio ആണ് strain. Elastic limit-യിനുള്ളിൽ stress $\propto$ strain എന്നാണ് Hooke's law.

Stress-strain relation



Important formulas

$$\text{stress} = F/A$$

Stress
pascal

$$\text{strain} = \frac{\text{change}}{\text{original}}$$

Strain
no unit

$$\text{stress} \propto \text{strain}$$

Hooke's law
within elastic limit

$$Y =$$

$$\frac{\text{stress}}{\text{strain}}$$

Young modulus
Pa

Symbols and SI units

Symbol	Meaning	SI unit
F	force	N
A	area	m ²
Y	Young modulus	Pa
ΔL	extension	m
L	original length	m

Real-life examples

- A spring returns to its original length after small stretch.
- A steel wire elongates slightly under load.

Engineering applications

- Springs, bridges, cranes, machine frames, material testing.

Study tip / memory trick: Stress has unit; strain has no unit.

Quick revision: Read the definition, formula, SI unit and one application. Most short-answer questions come from these four items.

MODULE 4 • LESSON 2

Pressure and surface tension

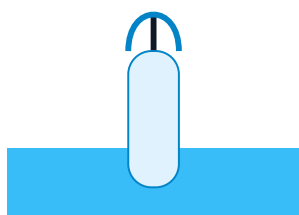
ENGLISH

Pressure is normal force per unit area. Atmospheric pressure is pressure due to air. Gauge pressure is pressure above atmospheric pressure, while absolute pressure equals atmospheric pressure plus gauge pressure. Surface tension is the property of liquid surface to behave like a stretched membrane.

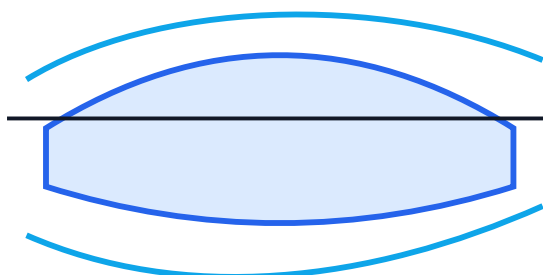
MALAYALAM

Unit area-ൽ normal ആയി acting ചെയ്യുന്ന force ആണ് pressure. Air കാരണം ഉണ്ടാകുന്ന pressure atmospheric pressure. Atmospheric pressure-നേക്കാൾ അധികമുള്ള pressure gauge pressure. Absolute pressure = atmospheric pressure + gauge pressure. Liquid surface stretched membrane പോലെ behave ചെയ്യുന്നതാണ് surface tension.

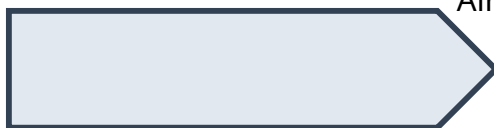
Fluid ideas



Capillary rise



Airfoil: faster air above → lower pressure



Important formulas

$$P = F/A$$

Pressure
Pa

$$P_{\text{abs}} = P_{\text{atm}} + P_{\text{gauge}}$$

Absolute pressure
Pa

$$h = \frac{2T \cos \theta}{r \rho g}$$

Ascent formula
capillary rise

Symbols and SI units

Symbol	Meaning	SI unit
P	pressure	Pa
T	surface tension	N m ⁻¹
θ	angle of contact	degree
r	capillary radius	m
ρ	density	kg m ⁻³

Real-life examples

- Needles can float on water due to surface tension.
- Capillary rise helps ink flow in narrow spaces.

Engineering applications

- Lubrication, detergents, painting, capillary tubes, pressure gauges.

Study tip / memory trick: Smaller area gives larger pressure for same force.

Quick revision: Read the definition, formula, SI unit and one application. Most short-answer questions come from these four items.

MODULE 4 • LESSON 3

Viscosity, terminal velocity and Stoke's law

ENGLISH

Viscosity is the internal friction of a fluid. A viscous fluid resists relative motion between its layers. Terminal velocity is the constant maximum velocity attained by a falling body in a fluid when net force becomes zero. Stoke's law gives viscous force on a small sphere moving slowly through a viscous fluid.

MALAYALAM

Fluid layers തമ്മിലുള്ള internal friction ആണ് viscosity. Viscous fluid layer motion-നെ oppose ചെയ്യും. ഒരു small body fluid-ൽ fall ചെയ്യുമ്പോൾ net force zero ആകുമ്പോൾ ലഭിക്കുന്ന constant maximum velocity terminal velocity ആണ്. Slow motion ചെയ്യുന്ന small sphere-ന് Stoke's law viscous force നൽകുന്നു.

Important formulas

$$F = 6\pi\eta r v$$

Stoke's law
viscous force

$$\eta \text{ unit} = \text{N s m}^{-2} \text{ or Pa s}$$

Coefficient unit
SI unit

Symbols and SI units

Symbol	Meaning	SI unit
η	coefficient of viscosity	Pa s
r	radius of sphere	m
v	velocity	m s ⁻¹
F	viscous force	N

Real-life examples

- Honey flows slower than water because viscosity is high.
- Rain drops attain terminal velocity.

Engineering applications

- Lubricating oils, hydraulic fluids, ball bearings, fluid transport pipelines.

Study tip / memory trick: More viscosity means more resistance to flow.

Quick revision: Read the definition, formula, SI unit and one application. Most short-answer questions come from these four items.

MODULE 4 • LESSON 4

Hydrodynamics, continuity equation and Bernoulli theorem

ENGLISH

Hydrodynamics deals with fluids in motion. In streamline flow, fluid particles follow smooth paths; in turbulent flow the motion is irregular. Reynolds number helps predict flow type. Continuity equation states that for incompressible fluid, area $\times$ velocity is constant. Bernoulli theorem states that pressure energy, kinetic energy and potential energy per unit volume remain constant along a streamline for ideal flow.

MALAYALAM

Motion ഉള്ള fluids പഠിക്കുന്നത് hydrodynamics ആണ്. Streamline flow smooth paths follow ചെയ്യും; turbulent flow irregular ആണ്. Flow type predict ചെയ്യാൻ Reynolds number ഉപയോഗിക്കുന്നു. Incompressible fluid-ൽ area $\times$ velocity constant എന്നാണ് continuity equation. Ideal streamline flow-ൽ pressure energy, kinetic energy, potential energy per unit volume total constant എന്നാണ് Bernoulli theorem.

Important formulas

$$Re = \frac{\rho v d}{\eta}$$

Reynolds number
no unit

$$A_1 v_1 = A_2 v_2$$

Continuity
incompressible flow

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

Bernoulli
along streamline

Symbols and SI units

Symbol	Meaning	SI unit
Re	Reynolds number	no unit
A	cross-sectional area	m ²
v	fluid velocity	m s ⁻¹
P	pressure	Pa
ρ	density	kg m ⁻³

Real-life examples

- Water flows faster at a narrow section of pipe.
- Atomiser works because high speed air lowers pressure and draws liquid up.

Engineering applications

- Venturimeter, atomiser, airfoil, carburetor, pipe networks and aircraft wings.

Study tip / memory trick: Narrow pipe $\rightarrow$ high velocity $\rightarrow$ lower pressure in Bernoulli effect.

Quick revision: Read the definition, formula, SI unit and one application. Most short-answer questions come from these four items.

Module 4 - Solved Numerical Problems

Every numerical follows the exam-friendly method: Given Data → Formula Used → Substitution → Calculation → Final Answer.

1. Stress in a wire

Given Data: Force $F = 1000 \text{ N}$, area $A = 2 \times 10^{-4} \text{ m}^2$.

Formula Used: stress = F/A

Substitution: stress = $1000 / (2 \times 10^{-4})$

Calculation Steps: stress = $5 \times 10^6 \text{ Pa}$.

Final Answer: Stress = 5 MPa

Shortcut Method: Divide by area in m^2 .

Common Mistakes: Using cm^2 without conversion.

2. Strain in a wire

Given Data: Original length $L = 2 \text{ m}$, extension $\Delta L = 0.001 \text{ m}$.

Formula Used: strain = $\Delta L/L$

Substitution: strain = $0.001/2$

Calculation Steps: strain = $0.0005 = 5 \times 10^{-4}$.

Final Answer: Strain = 5×10^{-4}

Shortcut Method: Strain has no unit.

Common Mistakes: Writing metre as strain unit.

3. Young's modulus

Given Data: Stress = 5×10^6 Pa, strain = 5×10^{-4} .

Formula Used: $Y = \text{stress}/\text{strain}$

Substitution: $Y = (5 \times 10^6)/(5 \times 10^{-4})$

Calculation Steps: $Y = 1 \times 10^{10}$ Pa.

Final Answer: Young's modulus = 1×10^{10} Pa

Shortcut Method: Cancel 5 and subtract powers.

Common Mistakes: Dividing strain by stress.

4. Absolute pressure

Given Data: Atmospheric pressure = 1.01×10^5 Pa, gauge pressure = 2×10^5 Pa.

Formula Used: $P_{\text{abs}} = P_{\text{atm}} + P_{\text{gauge}}$

Substitution: $P_{\text{abs}} = 1.01 \times 10^5 + 2 \times 10^5$

Calculation Steps: $P_{\text{abs}} = 3.01 \times 10^5$ Pa.

Final Answer: Absolute pressure = 3.01×10^5 Pa

Shortcut Method: Add gauge pressure to atmospheric pressure.

Common Mistakes: Confusing gauge pressure with absolute pressure.

5. Continuity equation

Given Data: $A_1 = 0.04 \text{ m}^2$, $v_1 = 2 \text{ m/s}$, $A_2 = 0.01 \text{ m}^2$.

Formula Used: $A_1 v_1 = A_2 v_2$

Substitution: $v_2 = A_1 v_1 / A_2 = 0.04 \times 2 / 0.01$

Calculation Steps: $v_2 = 0.08 / 0.01 = 8 \text{ m/s}$.

Final Answer: Velocity in narrow section = 8 m/s

Shortcut Method: If area becomes one-fourth, velocity becomes four times.

Common Mistakes: Forgetting inverse relation between area and velocity.

6. Capillary rise

Given Data: $T = 0.072 \text{ N/m}$, $\theta = 0^\circ$, $r = 0.5 \text{ mm} = 5 \times 10^{-4} \text{ m}$, $\rho = 1000 \text{ kg/m}^3$, $g = 9.8 \text{ m/s}^2$.

Formula Used: $h = \frac{2T \cos\theta}{r\rho g}$

Substitution: $h = \frac{2 \times 0.072 \times 1}{(5 \times 10^{-4} \times 1000 \times 9.8)}$

Calculation Steps: $h = 0.144/4.9 = 0.0294 \text{ m}$.

Final Answer: Capillary rise $\approx 2.94 \text{ cm}$

Shortcut Method: For water in clean glass, $\cos\theta \approx 1$.

Common Mistakes: Not converting mm to metre.

Module 4 - Practice Numericals with Answer Key

No.	Question	Answer / Hint
1	Force 500 N acts on area 0.01 m^2 . Find pressure.	50000 Pa
2	A wire of length 1 m extends by 0.001 m. Find strain.	0.001
3	Stress is $2 \times 10^6 \text{ Pa}$ and strain is 1×10^{-4} . Find Young modulus.	$2 \times 10^{10} \text{ Pa}$
4	Gauge pressure is $5 \times 10^4 \text{ Pa}$. Atmospheric pressure is $1 \times 10^5 \text{ Pa}$. Find absolute pressure.	$1.5 \times 10^5 \text{ Pa}$
5	Pipe area changes from 0.02 m^2 to 0.005 m^2 . If $v_1 = 1 \text{ m/s}$, find v_2 .	4 m/s
6	Find Reynolds number for $\rho = 1000 \text{ kg/m}^3$, $v = 0.2 \text{ m/s}$, $d = 0.01 \text{ m}$, $\eta = 0.001 \text{ Pa s}$.	2000
7	Find viscous force for $\eta = 0.5 \text{ Pa s}$, $r = 0.01 \text{ m}$, $v = 2 \text{ m/s}$. Use $\pi = 3.14$.	0.1884 N
8	If the same force acts on half the area, what happens to pressure?	Pressure becomes double

Numerical shortcut rule: Convert all quantities to SI units before substitution. Keep formula visible until the final step.

Module 4 - Exam-Oriented Questions and Answers

Short answer questions

Q1. Define Hooke's law.

Answer: Within elastic limit, stress is directly proportional to strain.

Q2. What is surface tension?

Answer: The force per unit length acting along the surface of a liquid.

Q3. What is terminal velocity?

Answer: The constant maximum velocity of a body falling through a fluid.

Q4. State continuity equation.

Answer: For incompressible fluid, $A_1v_1 = A_2v_2$.

Q5. Name two applications of Bernoulli theorem.

Answer: Airfoil and atomiser.

Viva-style quick questions

Q1. Viva: Define Hooke's law.

Answer: Within elastic limit, stress is directly proportional to strain.

Q2. Viva: What is surface tension?

Answer: The force per unit length acting along the surface of a liquid.

Q3. Viva: What is terminal velocity?

Answer: The constant maximum velocity of a body falling through a fluid.

Q4. Viva: State continuity equation.

Answer: For incompressible fluid, $A_1v_1 = A_2v_2$.

Q5. Viva: Name two applications of Bernoulli theorem.

Answer: Airfoil and atomiser.

Module 4 - Objective Practice with Answers

One-mark questions

Q1. Stress = force / ____.

Answer: area

Q2. Strain has unit/no unit?

Answer: No unit

Q3. Surface tension SI unit is ____.

Answer: N/m

Q4. Continuity equation is ____.

Answer: $A_1V_1 = A_2V_2$

Fill in the blanks

Q1. Young modulus = stress / ____.

Answer: strain

Q2. Absolute pressure = atmospheric pressure + ____.

Answer: gauge pressure

Q3. Stoke's law: $F =$ ____.

Answer: $6\pi\eta rv$

Q4. Bernoulli equation contains pressure, kinetic and ____ energy terms.

Answer: potential

True / False

Q1. Surface tension decreases with increase in temperature generally.

Answer: True

Q2. Pressure increases when area increases for same force.

Answer: False

Q3. Bernoulli theorem applies to ideal streamline flow.

Answer: True

Q4. Atomiser is an application of Bernoulli theorem.

Answer: True

Match the following

Item	Correct match
Stress	Pa
Viscosity	Pa s
Reynolds number	no unit
Bernoulli	airfoil

Assertion-Reason Questions

Q1. Assertion: Water flows faster in the narrow part of a pipe. Reason: $A_1v_1 = A_2v_2$.

Answer: Both true and reason explains assertion.

Q2. Assertion: Airfoil produces lift. Reason: Faster air above gives lower pressure.

Answer: Both true and reason explains assertion.

Module 4 - Long Answer Writing Practice

State Bernoulli's theorem and explain airfoil and atomiser as applications.

Key points to remember:

- Define ideal streamline flow.
- State Bernoulli equation $P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$.
- Explain that higher velocity gives lower pressure when height change is small.
- Airfoil: air moves faster over curved upper surface, pressure above becomes lower, lift is produced.
- Atomiser: fast air over tube lowers pressure and liquid rises and sprays.
- Mention limitations: non-viscous, incompressible, steady flow.

Important diagram to draw: Draw airfoil with faster air above and atomiser with fast air jet and liquid tube.

Mark-wise answer writing hint: For 10 marks, write statement, equation, assumptions, and two neat diagrams.

Sample model answer: Bernoulli's theorem states that for steady, incompressible and non-viscous flow along a streamline, the sum of pressure energy, kinetic energy and potential energy per unit volume remains constant. Therefore $P + \frac{1}{2}\rho v^2 + \rho gh$ is constant. If height change is small, an increase in velocity produces a decrease in pressure. In an airfoil, air flows faster over the curved upper surface than below it, so pressure above becomes lower than below. This pressure difference produces lift. In an atomiser, a high speed air stream passes over the top of a tube dipped in liquid. The pressure near the tube opening decreases and liquid rises and breaks into fine droplets.

1 mark

Write definition/formula with unit.

2 marks

Definition + one explanation or example.

5 marks

Statement + formula + diagram/application.

10 marks

Detailed explanation + derivation + diagram + application.

Module 4 - End-of-Module Revision Sheet

Quick formula cards

$$\text{stress} = F/A$$

Stress
pascal

$$\text{strain} = \text{change/original}$$

Strain
no unit

$$\text{stress} \propto \text{strain}$$

Hooke's law
within elastic limit

$$Y = \text{stress/strain}$$

Young modulus
Pa

$$P = F/A$$

Pressure
Pa

$$P_{\text{abs}} = P_{\text{atm}} + P_{\text{gauge}}$$

Absolute pressure
Pa

$$h = 2T \cos \theta / (r\rho g)$$

Ascent formula
capillary rise

$$F = 6\pi\eta r v$$

Stoke's law
viscous force

$$\eta \text{ unit} = \text{N s m}^{-2} \text{ or Pa s}$$

Coefficient unit
SI unit

$$Re = \rho v d / \eta$$

Reynolds number
no unit

$$A_1 v_1 = A_2 v_2$$

Continuity
incompressible flow

$$P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$$

Bernoulli
along streamline

Common mistakes to avoid

- Not writing SI units in final answer.
- Using formula without converting cm, mm or rpm to SI.
- Forgetting vector direction or sign.
- Skipping diagram in long answers.

Before exam checklist

- Can I define every key term?
- Can I write formulas from memory?
- Can I solve at least five numerical problems?
- Can I draw the important diagram neatly?

All exercise questions in this module include answers, answer keys, model answers or writing hints.

Final Master Answer Key - Practice Numericals

This section collects the answer keys from all modules for quick checking during revision.

Module 1

No.	Answer / Hint
1	12 N
2	6 kg m/s
3	10 N s
4	10 N
5	0.5 m
6	2%
7	24 kg m/s
8	10 m/s

Module 2

No.	Answer / Hint
1	2 m/s
2	5 m/s ²
3	32 N
4	2 kg m ² /s
5	0.4 m
6	0.5 m
7	6 rad/s
8	0.1

Module 3

No.	Answer / Hint
1	120 J
2	25 J
3	400 J
4	100 W
5	212°F
6	27°C
7	21000 J
8	20 N

Module 4

No.	Answer / Hint
1	50000 Pa
2	0.001
3	2×10^{10} Pa
4	1.5×10^5 Pa
5	4 m/s
6	2000
7	0.1884 N
8	Pressure becomes double

Frequently Asked Questions

How should I start numerical problems?

First write Given Data with units. Then write the formula. Substitute only after converting to SI units.

Should Malayalam terms be used in exam?

Use English technical terms in the answer sheet. Malayalam explanations in this PDF are for understanding concepts.

Which diagrams are most important?

Vector resolution, rocket propulsion, banking of roads, moment of inertia shapes, heat transfer modes, stress-strain graph, capillary rise, continuity pipe, airfoil and atomiser.

How to prepare in one day?

Revise formula cards, solve all worked examples, read long-answer key points, and check objective questions.

Reference Learning Path

Use this PDF as the main student-friendly guide. For deeper theory, refer to standard physics textbooks mentioned in the syllabus such as NCERT Physics, Applied Physics TTTI publications, H.C. Verma and Halliday/Resnick/Walker.

Final reminder: Neat diagrams, correct formulas, proper SI units and step-by-step numerical presentation can improve exam marks significantly.